

CSC490 Assignment 1 – Project Landscape

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Interest Statements

Domain: Analyzing Problematic Multimodal Content on Social Media

This domain captures both our personal and technical interests as a team. Having grown up with video-based social media platforms, we have developed an awareness of just how fast fake content can spread. In certain cases, we have also fallen victim to fake content online and have been unsure on several occasions whether certain audio and video content captured of public figures is legitimate. As computer science students collectively interested in both computer vision and multimodal content analysis, we are excited by the possibility of helping ourselves and others more critically navigate fake and dishonestly AI-generated content.

Why do we care about this project?

Pratibha Thakur

I am excited to build a mobile application for the first time and also to apply the knowledge on LLMs and reinforcement learning I have gained from research opportunities towards finetuning a model for AI-generated content detection.

Vaidik Patel

In this project, I am excited to contribute my experience related to Machine Learning Infrastructure and Large Language Models (LLMs) to develop a project that has a meaningful impact on user wellbeing.

Hardik Shah

I am excited in working with Machine Learning infrastructure on cloud as well as contributing to the project my knowledge of working with machine learning for the robotics space.

Paarth Arya

I'm excited to work on the technical challenge of turning raw video and text into features we can analyze for signs of falseness, while learning from existing approaches.

Landscape Analysis

This table is organized by category: ingesting data, extracting and chunking signals, leveraging datasets, detecting deepfakes and verifying content, grounding decisions with research.

Relevant Item	Description	Commentary
TikTok for Developers APIs	First-party endpoints for accessing public TikTok content (video metadata, captions, basic stats; aggregates).	Best path to a policy-compliant MVP. Access is gated and may not expose everything we want; we will use data uploads as a fallback.
OpenAI Whisper	Open-source speech-to-text for transcription from TikTok audio.	Allows claim detection and fact-checking. Compute-heavy but quantizable, batch-friendly.
PaddleOCR	Open-source OCR to extract overlaid/scene text from frames (captions, signs, lower-thirds).	Complements Whisper as input ingestion for text baked into the video; accuracy drops on stylized fonts/motion blur.
TransNet V2	Deep model to chunk short videos into coherent scenes.	Scene-level analysis reduces noise and helps align text/audio/visual cues.
FaceForensics++ Dataset	Open-source benchmark dataset of manipulated videos and accompanying deepfake detection framework.	Focused on face-centric deepfakes (does not cover audio fakes or non-face video falsifications).
ClaimBuster API	Scores transcript text to highlight “check-worthy” statements to verify.	Needs pairing with retrieval/fact-checks; occasional false positives acceptable for triage.
Google Fact Check Explorer API	Retrieves fact-checks from reputable outlets for a given claim.	Instantly flags debunked claims. Coverage is limited to previously checked statements.
sensity.ai	Commercial deepfake detection (upload URLs/files for multi-layer authenticity assessment).	Boasts 95% accuracy. Strong for face/voice manipulations. Not a fact-checker and not optimized for cross-video pattern analysis.
dfdc_deepfake_challenge	Baseline code + process for Facebook Deepfake Detection Challenge dataset (preprocess/train/eval).	Solid starting point for face-swap detection and evaluations. Research-oriented; not tailored to short-form UX or streaming.
Nasser et al. (2025) “A systematic review of multimodal fake news detection on social media using deep learning.”	Survey of deep-learning architectures/datasets for multimodal fake news on social media; highlights gaps (explainable AI, streaming).	Validates our multimodal direction and reminds us to invest in explainability and stream processing; a map for model choices and metrics.
Drolsbach & Pröllochs (2025) “Characterizing AI-Generated Misinformation on Social Media.”	Large-scale study of AI-generated misinformation on X (≈91k posts) analyzing patterns and risks.	Offers cues for exposure analytics and narrative repetition. Even though it’s X-text-centric, patterns translate to video captions/transcripts.

Project Outline

Problem Statement

Popular video-based social media platforms such as TikTok and Instagram often display hours of diverse video content to users daily. When fake videos with harmful or dishonestly AI-generated content surface in a young or adolescent user’s feed, an occurrence that is more likely for critical and high-impact topics (Drolsbach & Pröllochs, 2025), responses are suboptimal. Despite social media being one of young people’s preferred sources of news (Notley & Dezuanni, 2018), many ignore and disengage with potentially suspicious content (Selnes, 2024), rarely or never checking news sources (Notley & Dezuanni, 2018). Even young adults who rate their media literacy highly often “fail to identify the credibility of false, biased and vetted news” (Nygren & Guath, 2019).

Thus, the problem statement that we are interested in is this: **“How can we help users confidently and consciously consume and process information from video-based social media platforms?”**.

Proposed Solution

We propose a full-stack ML-powered mobile application designed to poll periodically for content consumed on TikTok and analyze high-risk video and audio for fake and dishonestly AI-generated content. The following table lists the requirements of our proposed solution.

Solution Requirements	
P0	Secure and legal polling for user content metadata
P0	Self-adapting AI-generated audio and video detection model
P0	Fake news detection model
P0	Dashboard on mobile application summarizing fake and dishonestly AI-generated
P0	LLM-generated descriptions explaining flagged content and providing context
P0	Mechanism for users to share feedback on flagged content
P1	Notification system to prompt users to check app once batch analysis completes
P1	Add capability to share a video directly from TikTok and get instant feedback
P1	Extend data-fetching to support analysis of Instagram Reels content
P1	Add support for analyzing patterns in consumed information
P1	Re-engineer user feeds by prompting users to revisit, flag, and remove content

High-Level Technical Approach (by Layer) and Milestones

User Data Input Layer: We will use the TikTok for Developers APIs with data uploads as fallback. We will collect videos, audio, and related metadata (captions, comments, hashtags). To collect data, we will create an ETL pipeline with cron jobs for periodic fetches.

Data Processing and Models Layer:

- **AI-generated Content Detection Model:** Fine-tuned multimodal models from HuggingFace for fake/AI-generated media. RLHF and topic-focused fine-tuning for short form, low resolution, and news related content (weak areas identified in literature). Scene extraction with TransNetV2 and video text extraction with PaddleOCR.
- **Fake News Detection Model:** ClaimsBuster API and Google Fact Check API with added RAG
- **Audio Analysis:** Speech transcription using OpenAI Whisper.
- **Content Safety:** Inappropriate content detection via Hive API / open-source alternatives.
- **LLM-Generated Explanations:** Prompt engineering workflow to generate user facing summaries that explain why content is flagged and provide contextualized background.
- **Feedback Loop for Model Improvements:** User input via thumbs up/down and occasional multiple-choice and A/B feedback prompts. Signals stored for future fine-tuning (closing the loop between users and models).

Storage Layer: Postgres for user content metadata, model outputs, and feedback.

Service Layer: FastAPI to aggregate model outputs and serve results to mobile clients.

Output Layer / User Interface: React Native mobile application summarizing flagged, AI-generated, or inappropriate content; exposure analysis of user feed.

The table below summarizes our project milestones.

Milestones	
Weeks 3-4	TikTok API ingestion pipeline; ETL for content aggregation; UI with mock data.
Weeks 4-5	Whisper audio transcription; initial HuggingFace deepfake detection models and fake news API setup (ClaimsBuster, Google Fact Check).
Weeks 5-7	Fine-tune multimodal detection model (visual and audio); add prompt engineering for generating user facing descriptions of detections; stand up cloud database and link to API.
Weeks 7-9	Integrate mobile app with backend services and deploy cloud infra.
Weeks 9-10	Load and stress testing of ingestion and processing pipelines; define and implement fault tolerance and error recovery policies.
Weeks 10-12	integrate all pipelines with UI; run complete user flow tests with mock and real TikTok data; deliver MVP demo with dashboard with detection results and feedback loop.

List of Unknowns

- Which available deepfake detection model is most suitable for fine-tuning to perform well on short-form, potentially blurry content related to critical topics in the news (war, health crises, and political tensions)?
- How well will the TikTok API work in practice? Is it brittle like other APIs (e.g. Meta's API for WhatsApp notification?) This will determine if we should fall back to user data uploads.
- How do we ensure strong privacy and transparency to maintain user trust?
- How do we naturally include the app in the user's flow of scrolling?

Press Release

New Mobile Application “Anchor” Detects Dishonest AI and Fake News on Your TikTok Feed

Toronto – December 5, 2025 – As Deepfakes become more sophisticated and fake news continues to go viral at unprecedented scale, Anchor empowers young social media users to detect harmful, misleading, or AI-generated fake content and gain transparency into their viewing habits.

The new mobile application, Anchor, cleanly flags and summarizes harmful, misleading, and dishonestly generated AI content that increasingly appears on our TikTok feeds. With the launch of this new application, young users can confidently consume news updates through a video-based social media platform, without fear of being misled or unknowingly spreading disinformation.

“Like most of my peers, I prefer getting news in short and digestible videos. TikTok gives me that, but everything going on with AI leaves me unsure about whether what I’m seeing or hearing is even real”, explains freshman Queens University student Alice Hamilton. “This application helps me verify the content I’ve consumed and feel more confident about the media I engage with”, says Alice.

In the background, Anchor uses state-of-the-art models finetuned on critical topics that recur on the news and may frequently be dishonestly modified: health warnings and epidemics, natural disasters, protests, footage of crime and war scenes, political discussions, and more. Through in-depth analysis of video, audio, and text metadata for each news-related video on a TikTok feed, Anchor is able to return a faithful summary of incorrect or twisted information. “We’ve created a solution to a problem that we ourselves have faced, and are excited to launch it to the millions of other TikTok and social media users who want to quickly yet reliably stay updated on the world’s most important issues”, says the team of four University of Toronto developers behind Anchor. “Our mission is to give users, especially in the younger generation, the tools we wish we had to navigate the increasingly complex and evolving information landscape. Instead of telling people what to think, we just want to give them additional context to think for themselves.”

“Innovations like Anchor give us hope”, explains Andy Eriksson from The Media Learning Association (MLA), a non-profit based in Sweden. “We know that the narratives pushed by video-based social media platforms like TikTok have a serious impact, and trust that tools like Anchor will help a new generation take a leap towards strong media literacy”.

To get started, users can simply install the Anchor App from the iOS or Android app store, create an account, and securely link their TikTok in just a few taps. Anchor will have an in-depth analysis prepared every time users end a session.

Appendix: Iterations Considered

Iteration 1: Alternate TikTok-Style Mobile Application

- Concept: Build a full mobile application that serves as an alternative TikTok, integrating AI-powered detection directly into the feed. This new app would directly flag problematic or fake content.
- Strengths: Seamlessly integrated experience, aligns with real user behavior (mobile-first), offers full control over the feed and UI.
- Weaknesses: Very high development and maintenance overhead (essentially rebuilding TikTok's interface), possible legal/ToS issues.

Iteration 2: Browser Extension Prototype

- Concept: A lightweight Chrome/Firefox extension that passively scans TikTok or Instagram web sessions and highlights suspicious or AI-generated content.
- Strengths: Quick to build, easy user adoption (no app install), lower overhead.
- Weaknesses: Limited to browser users (most TikTok/IG consumption is mobile), API/data access through extensions is restricted and brittle.

Iteration 3: Exposure Pattern Analysis Focus

- Concept: Rather than flagging individual videos, focus on analyzing aggregate viewing patterns (bias, echo chambers, dark patterns). Insights would be shown in dashboards or summaries, helping users reflect on their content diet.
- Strengths: Novel and research-aligned contribution, improves media literacy, scalable to multiple platforms.
- Weaknesses: Less immediate feedback for users (doesn't block or flag content directly), requires larger datasets and longitudinal tracking.

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